GeneSight® Psychotropic

Pharmacogenomic Test

genesight

Attawar, Vrijen
Date of Birth: 11/12/1992
Clinician: Shirin Ali MD

Order Number: 4150734
Report Date: 2/23/2022
Reference:

Questions about report interpretation?
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855.891.9415 | medinfo@genesight.com

Antidepressants

Use as Directed	Moderate Gene-drug Interaction		Significant Gene-drug Interaction
amitriptyline (Elavil®)	**bupropion** (Wellbutrin®)	1	
clomipramine (Anafranil®)	**selegiline** (Emsam®)	1	
desipramine (Norpramin®)	**citalopram** (Celexa®)	4	
desvenlafaxine (Pristiq®)	**escitalopram** (Lexapro®)	4	
doxepin (Sinequan®)	**fluoxetine** (Prozac®)	4	
duloxetine (Cymbalta®)	**fluvoxamine** (Luvox®)	4	
imipramine (Tofranil®)	**paroxetine** (Paxil®)	4	
levomilnacipran (Fetzima®)	**sertraline** (Zoloft®)	4	
mirtazapine (Remeron®)			
nortriptyline (Pamelor®)			
trazodone (Desyrel®)			
venlafaxine (Effexor®)			
vilazodone (Viibryd®)			
vortioxetine (Trintellix®)			

Clinical Considerations

1: Serum level may be too high, lower doses may be required.
4: Genotype may impact drug mechanism of action and result in moderately reduced efficacy.

All psychotropic medications require clinical monitoring. Medications should not be changed based solely on the test results. The results of this test are intended to supplement other clinical information considered by a healthcare provider within the context of a comprehensive medical evaluation.
This report is not intended to imply that the drugs listed are approved for the same indications or that they are comparable in safety or efficacy. The brand name is shown for illustrative purposes only; other brand names may be available. The prescribing physician should review the prescribing information for the drug(s) being considered and make treatment decisions based on the patient's individual needs and the characteristics of the drug prescribed. Propranolol and oxcarbazepine might be considered off-label when being used for neuropsychiatric disorders. Please consult their respective FDA drug labels for specific guidelines regarding their use.

The GeneSight Psychotropic test interpretations are based on a thorough review of published peer-reviewed literature, internal research, and FDA label information when applicable. The clinical validity and utility of the GeneSight Psychotropic test have been evaluated for patients with major depressive disorder who failed at least one psychotropic medication in multiple clinical studies.

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Anxiolytics and Hypnotics

Use as Directed	Moderate Gene-drug Interaction		Significant Gene-drug Interaction
alprazolam (Xanax®)	**chlordiazepoxide** (Librium®)	1	
buspirone (BuSpar®)	**clorazepate** (Tranxene®)	1	
clonazepam (Klonopin®)	**diazepam** (Valium®)	1	
eszopiclone (Lunesta®)	**lorazepam** (Ativan®)	1	
propranolol (Inderal®)	**oxazepam** (Serax®)	1	
temazepam (Restoril®)			
zolpidem (Ambien®)			

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Antipsychotics

Use as Directed	Moderate Gene-drug Interaction	Significant Gene-drug Interaction
aripiprazole (Abilify®)		
asenapine (Saphris®)		
brexpiprazole (Rexulti®)		
cariprazine (Vraylar®)		
chlorpromazine (Thorazine®)		
clozapine (Clozaril®)		
fluphenazine (Prolixin®)		
haloperidol (Haldol®)		
iloperidone (Fanapt®)		
lumateperone (Caplyta®)		
lurasidone (Latuda®)		
olanzapine (Zyprexa®)		
paliperidone (Invega®)		
perphenazine (Trilafon®)		
quetiapine (Seroquel®)		
risperidone (Risperdal®)		
thioridazine (Mellaril®)		
thiothixene (Navane®)		
ziprasidone (Geodon®)		

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Mood Stabilizers

Use as Directed	Moderate Gene-drug Interaction	Significant Gene-drug Interaction
carbamazepine (Tegretol®) **lamotrigine** (Lamictal®) **oxcarbazepine** (Trileptal®) **valproic acid/divalproex** (Depakote®)		

No Proven Genetic Markers

gabapentin (Neurontin®)	10	**lithium** (Eskalith®)	10	**topiramate** (Topamax®)	10

Clinical Considerations

10: While this medication does not have clinically proven genetic markers that allow it to be categorized, it may be an effective choice based on other clinical factors.

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Stimulants

Use as Directed	Moderate Gene-drug Interaction	Significant Gene-drug Interaction
dexmethylphenidate (Focalin®) **methylphenidate** (Ritalin®, Concerta®)		

No Proven Genetic Markers

amphetamine salts (Adderall®)	10	**dextroamphetamine** (Dexedrine®)	10	**lisdexamfetamine** (Vyvanse®)	10

Non-Stimulants

Use as Directed	Moderate Gene-drug Interaction	Significant Gene-drug Interaction
atomoxetine (Strattera®) **guanfacine** (Intuniv®)		

No Proven Genetic Markers

clonidine (Kapvay®)	10

Clinical Considerations

10: While this medication does not have clinically proven genetic markers that allow it to be categorized, it may be an effective choice based on other clinical factors.

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Patient Genotypes and Phenotypes

Pharmacodynamic Genes

PD

ADRA2A **Normal Response**

C/G

This patient is heterozygous for the -1291G>C polymorphism in the adrenergic alpha-2A receptor gene. They have one copy of the C allele and one copy of the G allele. This genotype suggests a normal response to certain ADHD medications.

HLA-A*3101 **Lower Risk**

A/A

This patient is homozygous for the A allele of the rs1061235 A>T polymorphism indicating absence of the HLA-A*3101 allele. This genotype suggests a lower risk of serious hypersensitivity reactions, including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), maculopapular eruptions, and Drug Reaction with Eosinophilia and Systemic Symptoms when taking certain mood stabilizers.

HLA-B*1502 **Lower Risk**

Not Present

This patient does not carry the HLA-B*1502 allele or a closely related *15 allele. Absence of HLA-B*1502 and the closely related *15 alleles suggests lower risk of serious dermatologic reactions including toxic epidermal necrolysis (TEN) and Stevens-Johnson syndrome (SJS) when taking certain mood stabilizers.

HTR2A **Normal Sensitivity**

G/A

This individual is heterozygous for the G allele and A allele of the -1438G>A polymorphism for the Serotonin Receptor Type 2A. They have one copy of the G allele. This genotype is not predictive of adverse drug reactions with selective serotonin reuptake inhibitors.

SLC6A4 **Reduced Response**

S/S

This patient is homozygous for the short promoter polymorphism of the serotonin transporter gene. The short promoter allele is reported to decrease expression of the serotonin transporter compared to the homozygous long promoter allele. The patient may have a moderately decreased likelihood of response to selective serotonin reuptake inhibitors due to the presence of the short form of the gene.

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Patient Genotypes and Phenotypes

Pharmacokinetic Genes

PK

CES1A1 **Extensive (Normal) Metabolizer**

GLY/GLY

CES1A1 - Gly allele enzyme activity: Normal
CES1A1 - Gly allele enzyme activity: Normal

This genotype is most consistent with the extensive (normal) metabolizer phenotype. The patient is expected to have normal enzyme activity.

CYP1A2 **Extensive (Normal) Metabolizer**

-163C>A - C/A, 5347C>T - C/T

This genotype is most consistent with the extensive (normal) metabolizer phenotype.

CYP2B6 **Intermediate Metabolizer**

*1/*6

CYP2B6*1 allele enzyme activity: Normal
CYP2B6*6 allele enzyme activity: Reduced

This genotype is most consistent with the intermediate metabolizer phenotype. This patient may have reduced enzyme activity as compared to individuals with the normal phenotype.

CYP2C19 **Extensive (Normal) Metabolizer**

*2/*17

CYP2C19*2 allele enzyme activity: None
CYP2C19*17 allele enzyme activity: Increased

This genotype is most consistent with the extensive (normal) metabolizer phenotype.

CYP2C9 **Extensive (Normal) Metabolizer**

*1/*1

CYP2C9*1 allele enzyme activity: Normal
CYP2C9*1 allele enzyme activity: Normal

This genotype is most consistent with the extensive (normal) metabolizer phenotype.

CYP2D6 **Extensive (Normal) Metabolizer**

*1/*2A

CYP2D6*1 allele enzyme activity: Normal
CYP2D6*2A allele enzyme activity: Increased

This genotype is most consistent with the extensive (normal) metabolizer phenotype.

CYP3A4 **Extensive (Normal) Metabolizer**

*1/*1

CYP3A4*1 allele enzyme activity: Normal
CYP3A4*1 allele enzyme activity: Normal

This genotype is most consistent with the extensive (normal) metabolizer phenotype.

UGT1A4 **Extensive (Normal) Metabolizer**

*1/*1

UGT1A4*1 allele enzyme activity: Normal
UGT1A4*1 allele enzyme activity: Normal

This genotype is most consistent with the extensive (normal) metabolizer phenotype. The patient is expected to have normal enzyme activity.

UGT2B15 **Intermediate Metabolizer**

*2/*2

UGT2B15*2 allele enzyme activity: Reduced
UGT2B15*2 allele enzyme activity: Reduced

This genotype is most consistent with the intermediate metabolizer phenotype. This patient may have reduced enzyme activity as compared to individuals with the normal phenotype.

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Additional Genotypes

Not Included in Categorizing Medications

Genotypes reported in this section have not been shown to be reliable markers of medication outcomes

COMT

VAL/VAL

This patient is homozygous for the Val allele of the Val158Met polymorphism in the catechol-o-methyltransferase gene.

A summary of the studies that have assessed the potential effect of COMT genotype on response to psychotropic medications can be found here: https://genesight.com/comt

To categorize medications on this pharmacogenomic test, a gene must have a variant that has been shown to have a significant impact on medication outcomes, as demonstrated in multiple well-designed studies. Studies assessing the gene in this section have not shown that it is a reliable marker of medication outcomes. Therefore, this gene does not currently meet the criteria for categorizing medications. The patient's genotype is provided for informational purposes only.

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Gene-drug Interactions

Use as Directed

	CES1A1 Normal	CYP1A2 Normal	CYP2B6 Intermediate	CYP2C19 Normal	CYP2C9 Normal	CYP2D6 Normal	CYP3A4 Normal	UGT1A4 Normal	UGT2B15 Intermediate
Antidepressants									
amitriptyline (Elavil®)		○		○	○	○	○	○	
clomipramine (Anafranil®)		○		○		○	○		
desipramine (Norpramin®)						○			
desvenlafaxine (Pristiq®)				○			○		
doxepin (Sinequan®)		○		○	○	○	○	○	
duloxetine (Cymbalta®)		○				○			
imipramine (Tofranil®)		○		○		○	○		
levomilnacipran (Fetzima®)				○		○	○		
mirtazapine (Remeron®)		○			○	○	○		
nortriptyline (Pamelor®)						○			
trazodone (Desyrel®)		○				○	○		
venlafaxine (Effexor®)				○	○	○	○		
vilazodone (Viibryd®)				○		○	○		
vortioxetine (Trintellix®)			●	○	○	○	○		
Anxiolytics and Hypnotics									
alprazolam (Xanax®)							○		
buspirone (BuSpar®)						○	○		
clonazepam (Klonopin®)							○		
eszopiclone (Lunesta®)					○		○		
propranolol (Inderal®)		○				○			
temazepam (Restoril®)			●		○		○		●
zolpidem (Ambien®)		○		○	○	○	○		
Antipsychotics									
aripiprazole (Abilify®)						○	○		
asenapine (Saphris®)		○				○	○	○	
brexpiprazole (Rexulti®)						○	○		
cariprazine (Vraylar®)						○	○		
chlorpromazine (Thorazine®)		○				○	○		
clozapine (Clozaril®)		○				○	○	○	
fluphenazine (Prolixin®)		○		○	○	○	○		
haloperidol (Haldol®)		○				○	○	○	

● Variation was found in patient genotype that may impact medication metabolism.

○ This gene is associated with medication metabolism, but the predicted patient phenotype is normal.

* This gene-drug interaction is recognized by the FDA or CPIC.

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Gene-drug Interactions

Use as Directed (Continued)

	CES1A1 Normal	CYP1A2 Normal	CYP2B6 Intermediate	CYP2C19 Normal	CYP2C9 Normal	CYP2D6 Normal	CYP3A4 Normal	UGT1A4 Normal	UGT2B15 Intermediate
Antipsychotics									
iloperidone (Fanapt®)						○	○		
lumateperone (Caplyta®)							○		
lurasidone (Latuda®)							○		
olanzapine (Zyprexa®)		○				○	○	○	
paliperidone (Invega®)						○	○		
perphenazine (Trilafon®)		○		○		○	○		
quetiapine (Seroquel®)						○	○		
risperidone (Risperdal®)						○	○		
thioridazine (Mellaril®)		○		○		○	○		
thiothixene (Navane®)		○							
ziprasidone (Geodon®)		○					○		
Mood Stabilizers									
carbamazepine (Tegretol®)			●				○		
lamotrigine (Lamictal®)								○	
oxcarbazepine (Trileptal®)									
valproic acid/divalproex (Depakote®)			●		○			○	
Stimulants									
dexmethylphenidate (Focalin®)	○								
methylphenidate (Ritalin®, Concerta®)	○								
Non-Stimulants									
atomoxetine (Strattera®)						○			
guanfacine (Intuniv®)							○		

Moderate Gene-Drug Interaction

	CES1A1 Normal	CYP1A2 Normal	CYP2B6 Intermediate	CYP2C19 Normal	CYP2C9 Normal	CYP2D6 Normal	CYP3A4 Normal	UGT1A4 Normal	UGT2B15 Intermediate
Antidepressants									
bupropion (Wellbutrin®)			●			○	○		
citalopram (Celexa®)				○		○	○		

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Antidepressants									
escitalopram (Lexapro®)				○		○	○		
fluoxetine (Prozac®)				○	○	○	○		
fluvoxamine (Luvox®)		○				○			
paroxetine (Paxil®)						○	○		
selegiline (Emsam®)		○	●	○			○		
sertraline (Zoloft®)			●	○			○		
Anxiolytics and Hypnotics									
chlordiazepoxide (Librium®)		○					○		●
clorazepate (Tranxene®)		○					○		●
diazepam (Valium®)		○	●	○	○		○		●
lorazepam (Ativan®)									●
oxazepam (Serax®)									●

No Proven Genetic Markers

	CES1A1 Normal	CYP1A2 Normal	CYP2B6 Intermediate	CYP2C19 Normal	CYP2C9 Normal	CYP2D6 Normal	CYP3A4 Normal	UGT1A4 Normal	UGT2B15 Intermediate
Stimulants									
amphetamine salts (Adderall®)									

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Test Information

The buccal swab sample was collected on 2/17/2022 and received in the laboratory on 2/22/2022. Genomic DNA was isolated and the relevant genomic regions were amplified by polymerase chain reaction (PCR). Analysis of CYP2D6 deletion and duplication, HLA-B*1502 and SLC6A4 was completed by electrophoresis of PCR products. Analysis of rs1061235 (indicating presence of the HLA-A*3101 allele or certain HLA-A*33 alleles), ADRA2A, CES1A1, COMT, CYP1A2, CYP2B6, CYP2C19, CYP2C9, CYP2D6, CYP3A4, HTR2A, UGT1A4 and UGT2B15 was completed by using iPLEX MassARRAY® technology (Agena Bioscience). The following genetic variants may be detected in the assay: ADRA2A -1291C>G (NM_000681.3:c.-1252C>G); CES1A1 (NM_001025194.1:c.428G>A); COMT Val158Met (NM_007310.2:c.322G>A); CYP1A2 -3860G>A (NG_008431.1:g.28338G>A), -2467T>delT (NM_000761.4:c.-1635delT), -739T>G (NM_000761.4:c.-10+103T>G), -729C>T (NM_000761.4:c.-10+113C>T), -163C>A (NM_000761.4:c.-9-154C>A), 125C>G (NM_000761.4:c.125C>G), 558C>A (NM_000761.4:c.558C>A), 2116G>A (NM_000761.4:c.1042G>A), 2473G>A (NM_000761.4:c.1130G>A), 2499A>T (NM_000761.4:c.1156A>T), 3497G>A (NM_000761.4:c.1217G>A), 3533G>A (NM_000761.4:c.1253+1G>A), 5090C>T (NM_000761.4:c.1291C>T), 5166G>A (NM_000761.4:c.1367G>A), 5347C>T (NM_000761.4:c.1548C>T); CYP2B6 *1, *4 (NM_000767.4:c.785A>G), *6 (NM_000767.4:c.516G>T; c.785A>G), *9 (NM_000767.4:c.516G>T); CYP2C19 *1, *2 (NM_000769.2:c.681G>A), *3 (NM_000769.2:c.636G>A), *4 (NM_000769.2:c.1A>G), *5 (NM_000769.2:c.1297C>T),*6 (NM_000769.2:c.395G>A), *7 (NM_000769.2:c.819+2T>A), *8 (NM_000769.2:c.358T>C), *17 (NM_000769.2:c.-806C>T); CYP2C9 *1, *2 (NM_000771.3:c.430C>T), *3 (NM_000771.3:c.1075A>C), *4 (NM_000771.3:c.1076T>C), *5 (NM_000771.3:c.1080C>G), *6 (NM_000771.3:c.817delA); CYP2D6 *1, *2 (NM_000106.5:c.886C>T; c.1457G>C), *2A (NM_000106.5:c.-1584C>G; c.886C>T; c.1457G>C), *3 (NM_000106.5:c.775delA), *4 (NM_000106.5:c.506-1G>A; c.100C>T; c.1457G>C), *5 (CYP2D6 Deletion), *6 (NM_000106.5:c.454delT), *7 (NM_000106.5:c.971A>C), *8 (NM_000106.5:c.505G>T; c.886C>T; c.1457G>C), *9 (NM_000106.5:c.841_843delAAG), *10 (NM_000106.5:c.100C>T; c.1457G>C), *11, *12 (NM_000106.5:c.124G>A; c.886C>T; c.1457G>C), *14 (NM_000106.5:c.505G>A; c.886C>T; c.1457G>C), *15, *17 (NM_000106.5:c.320C>T; c.886C>T; c.1457G>C), *41 (NM_000106.5:c.985+39G>A; c.886C>T; c.1457G>C), gene duplication; CYP3A4 *1, *13 (NM_017460.5:c.1247C>T), *15A (NM_017460.5:c.485G>A), *22 (NM_017460.5:c.522-191C>T); HLA-B*1502; rs1061235 (NM_002116.7:c.*66A>T); HTR2A -1438G>A (NM_000621.4:c.-998G>A); SLC6A4 L, S; UGT1A4 *1, *3 (NM_007120.2:c.142T>G); UGT2B15 *1, *2 (NM_001076.3:c.253G>T). The following rare genetic variants have not been observed by the Assurex Health, Inc. laboratory: CYP1A2 125C>G, 558C>A; CYP2C19 *7. *1 is the reference allele and is reported by default if the other tested alleles are not detected.

This test was developed and its performance characteristics determined by Assurex Health. It has not been cleared or approved by the U.S. Food and Drug Administration. These interpretations are based upon data available in scientific literature and prescribing information for the relevant drugs. Interpretations are, in some instances, based on data regarding the pharmacokinetic, pharmacodynamic and pharmacogenomics properties of a drug derived from non-clinical studies (e.g. *in vitro* studies). Findings from studies performed in a non-clinical setting or clinical studies involving healthy subjects are not necessarily indicative of clinical performance in a particular patient. References used to inform medication categorizations can be found here: https://genesight.com/references.

This report was reviewed and verified on 2/23/2022 by:

Bailing Li, PhD

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The information contained in this report is provided as a service and does not constitute medical advice. At the time of report generation this information is believed to be current and is based upon published research; however, research data evolves and amendments to the prescribing information of the drugs listed will change over time. While this report is believed to be accurate and complete as of the date issued, THE DATA IS PROVIDED "AS IS", WITHOUT WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT LIMITATION, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. As medical advice must be tailored to the specific circumstances of each case, the treating healthcare provider has ultimate responsibility for all treatment decisions made with regard to a patient including any made on the basis of a patient's genotype.

GeneSight Psychotropic is covered by U.S. Patent No. 9,111,028

Genetic testing was completed by a CLIA and CAP accredited laboratory in the United States located at:
6000 Mason-Montgomery Road
Mason, OH 45040
Laboratory Director: Nina King, PhD

Customer Service

Please contact 855.891.9415 or medinfo@genesight.com for assistance with report interpretation. For all other inquires please contact 866.757.9204 or support@genesight.com.

GeneSight Psychotropic Test Version: 4.0

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GeneSight® MTHFR
COMBINATORIAL PHARMACOGENOMIC TEST

genesight

Attawar, Vrijen
DOB: 11/12/1992
Order Number: 4150734
Report Date: 2/23/2022
Clinician: Shirin Ali MD
Reference:

Questions about report interpretation?
Contact our Medical Information team
855.891.9415
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NORMAL FOLIC ACID CONVERSION	REDUCED FOLIC ACID CONVERSION	SIGNIFICANTLY REDUCED FOLIC ACID CONVERSION
	✓	

Note: Serum levels of folate may be too low. Folate supplementation or higher daily intake of folic acid may be required.

PATIENT GENOTYPE AND PHENOTYPE

MTHFR	Intermediate Activity	C/T

This individual is heterozygous for the C677T polymorphism in the MTHFR gene. This genotype is associated with reduced folic acid metabolism, moderately decreased serum folate levels, and moderately increased homocysteine levels.

TEST INFORMATION

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